SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

NAME: K. HEMA SAI SIDDARTHA

DATE:08/10/2026

LAB EXP: 3

Reg Mo: 192325036

Course Code/ Name: DSA0601 / Data Handling and Visualization

Aim

To analyze geographic data of three cities using a map chart, scatter plot and table, and to build an interactive Tableau dashboard for exploring the geographic data.

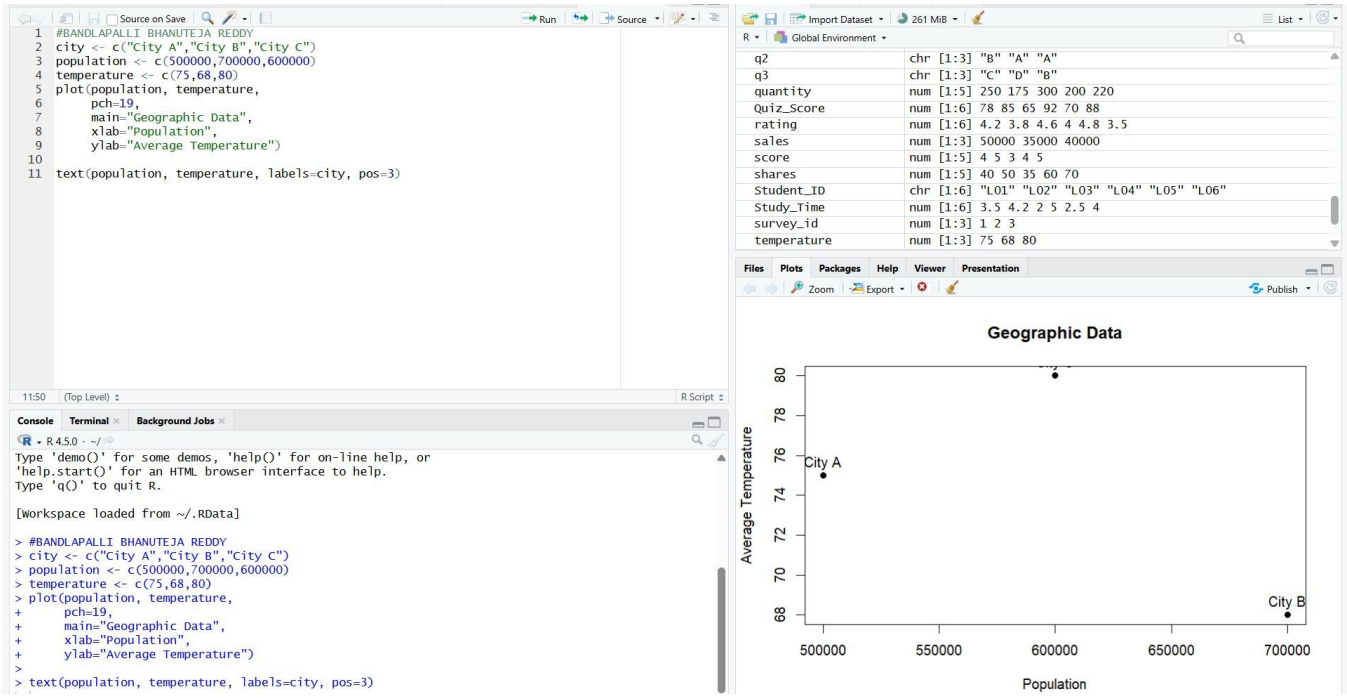
Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Question 1: Map Chart of Cities

Question: Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

Aim: To show the three cities on a geographic map using labelled markers sized by population.

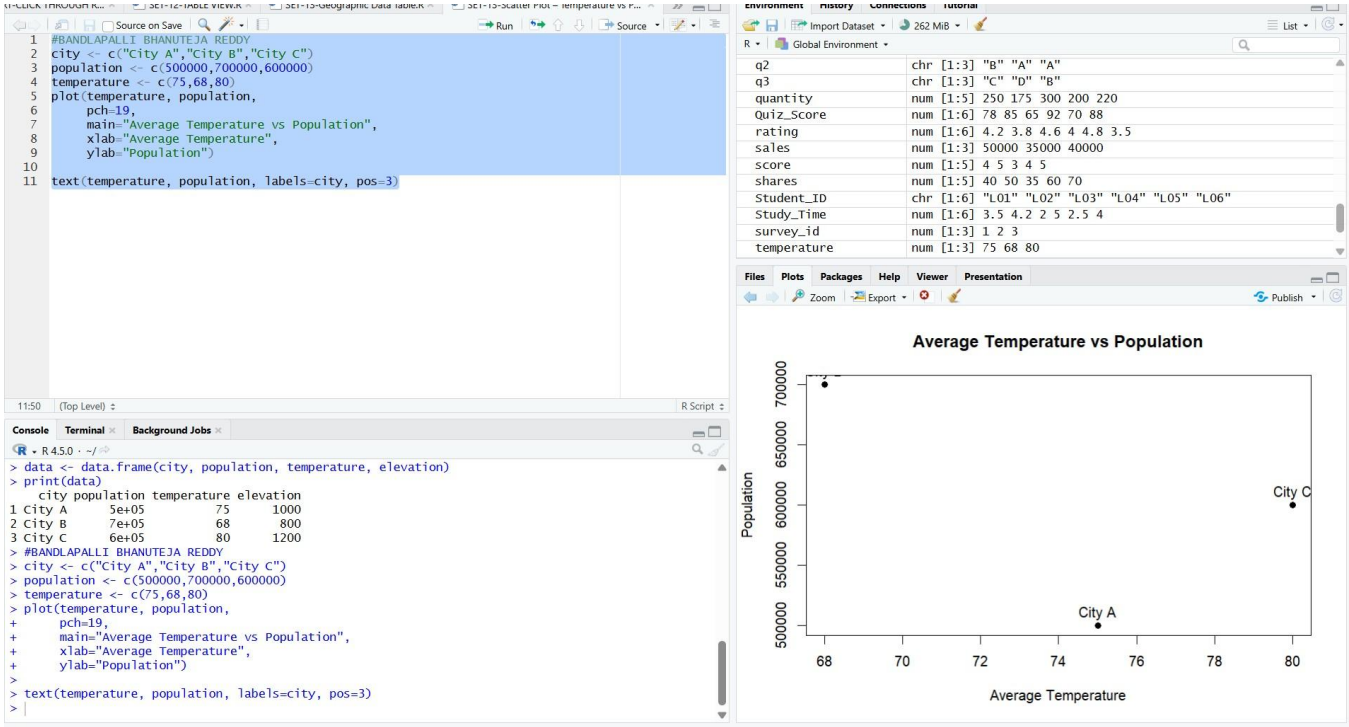


Result: City B has the largest marker (700000), followed by City C (600000) and City A (500000); the dataset has no latitude or longitude, so the real locations must be supplied.

Question 2: Scatter Plot of Temperature vs Population

Question: Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

Aim: To explore how average temperature and population size are related using a scatter plot.

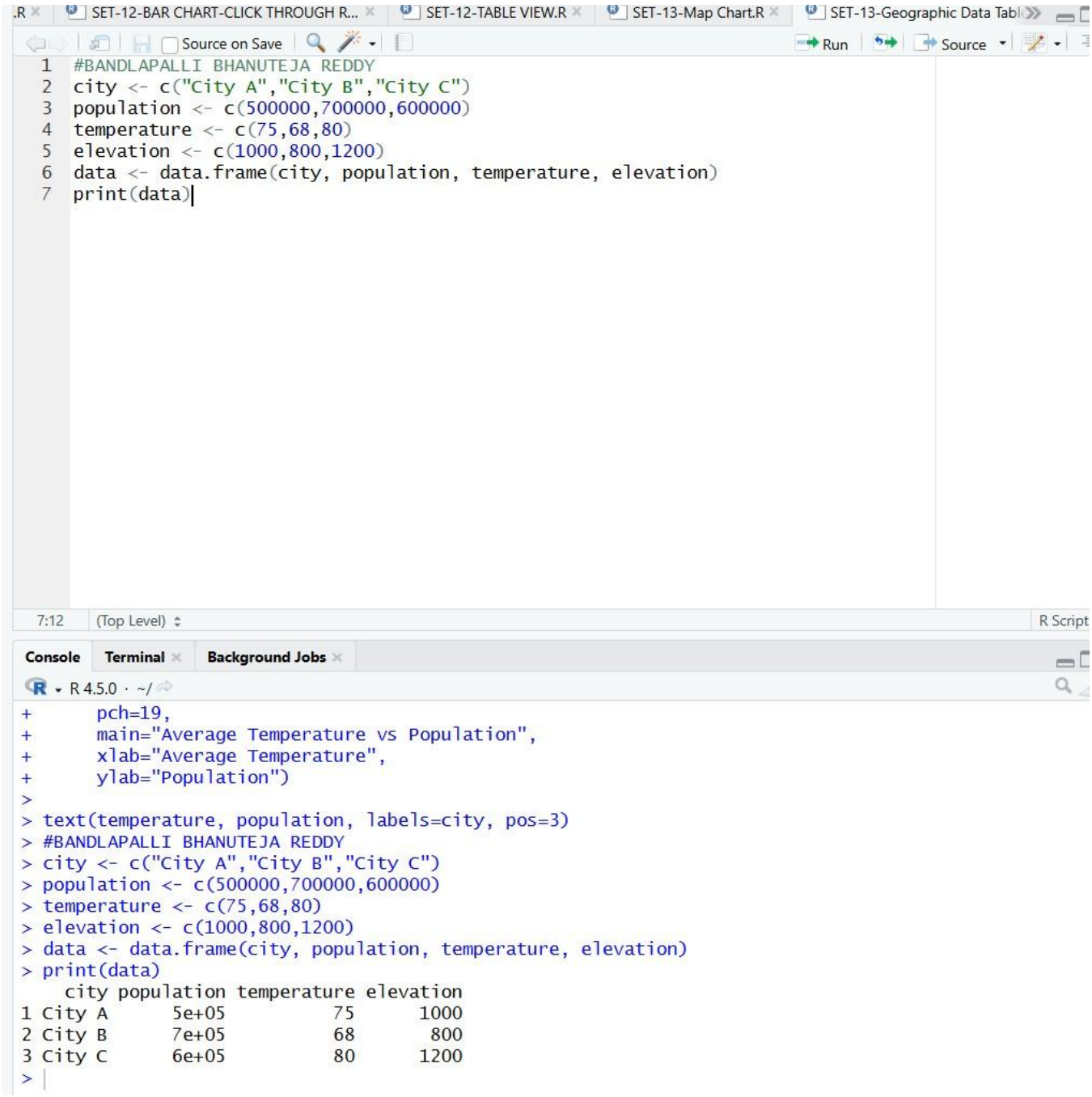


Result: The correlation is about -0.58, a moderate negative relationship, because the coolest city (City B, 68) has the highest population; with only three cities this is only an indication.

Question 3: Table of Geographic Data

Question: Build a table to display the geographic data for each city. Label the table elements.

Aim: To present the population, average temperature and elevation of each city in a clearly labelled table.

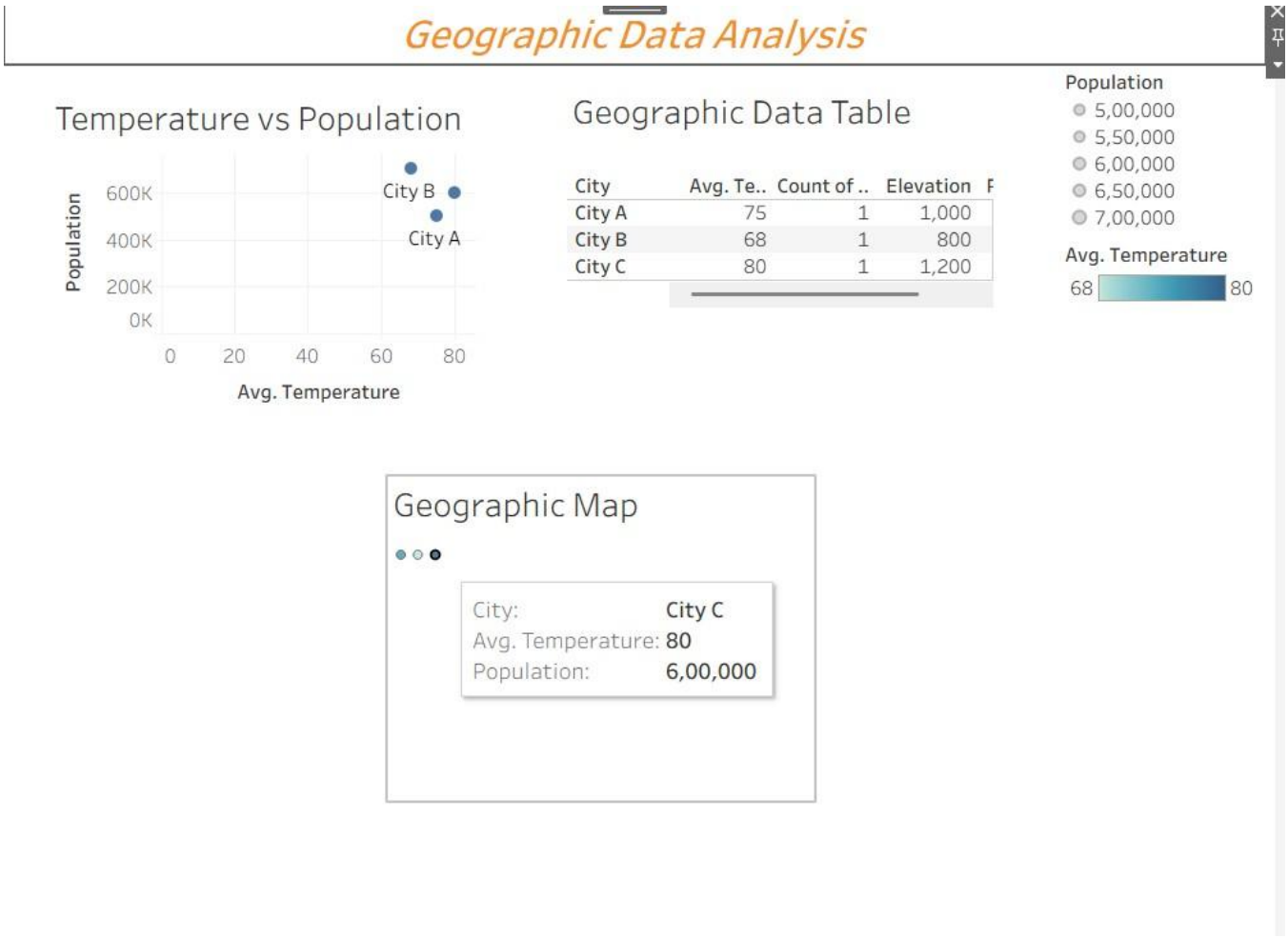


Result: The table lists all three cities; City C is the hottest (80) and the highest (1200), while City B has the largest population and the lowest elevation (800).

Question 4: Interactive Tableau Dashboard

Question: Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Aim: To build an interactive Tableau dashboard that combines the map chart, scatter plot and table for exploring geographic data.



Result: The dashboard places all three views together, and selecting a city on the map highlights it in the scatter plot and the table.

Overall Result

Observation: The three cities have a total population of 1,800,000, an average temperature of about 74.3 and an average elevation of 1000.